

Lesson plan

- ① **Introduction:** Painting the big picture for ensemble DA methods.
- ② **Stochastic EnKFs:** Presenting the original EnKF algorithm in which each forecast member is paired with a realization of the observation errors.
- ③ **Deterministic EnKFs:** Formulating another family of EnKF methods which avoid the need to simulate from the observation error distribution.

Introduction

- Q *Stochastic EnKFs are required to sample from the distribution of observation errors $\mathbf{E}_k^o$. Why is this not always desirable?* *Introduce additional sampling error.*
- As their name suggests, deterministic (or square-root) EnKFs circumvent this issue. However, we already saw that simply omitting the observation error samples in the stochastic EnKF update results in the underestimation of the KF's analysis covariance.
- Therefore, deterministic EnKFs look for an **alternative linear transformation**

$$\left\{ \mathbf{x}_k^{f,i} \right\}_{i=1}^{N_e} \mapsto \left\{ \mathbf{x}_k^{a,i} \right\}_{i=1}^{N_e}$$

which is *deterministic in its inputs but consistent with the KF update.*

Analysis mean I

- Here we outline 1 possible formulation in order to focus our attention on the general procedure. In upcoming lectures, we will review other algorithmic variants.
- The first step in deterministic EnKFs is to update the **forecast mean** $\mathbf{x}^f \approx \overline{\mathbf{X}}^f = N_e^{-1} \sum_{i=1}^{N_e} \mathbf{X}_i^f$:

$$\overline{\mathbf{X}}^a = \overline{\mathbf{X}}^f + \hat{\mathbf{K}}(\mathbf{y} - \mathbf{H}\overline{\mathbf{X}}^f).$$

- In the stochastic EnKF, we had that $\hat{\mathbf{K}} = \mathbf{X}\mathbf{Y}^\top(\mathbf{Y}\mathbf{Y}^\top)^{-1}$ with

$$\mathbf{Y} = \text{matrix with columns } (N_e - 1)^{-1/2} \left[\mathbf{H}\mathbf{X}_i^f + \mathbf{E}_i^o - (\mathbf{H}\overline{\mathbf{X}}^f + \overline{\mathbf{E}}^o) \right].$$

- However, this approximation is not appropriate as we want to **avoid sampling the observation errors $\mathbf{E}^o$** .

Analysis mean II

- To do so, we resort to the same **trick** needed to avoid sampling noise in the observation error ensemble $\{\mathbf{E}_i^o\}_{i=1}^{N_e}$ (i.e., the possibility for $\bar{\mathbf{E}}^o \neq \mathbf{0}$) – **explicitly using R**.
- Recall that the Kalman gain is given by $\mathbf{K} = \mathbf{P}^f \mathbf{H}^\top (\mathbf{H} \mathbf{P}^f \mathbf{H}^\top + \mathbf{R})^{-1}$. Using our ensemble approximation $\mathbf{P}^f \approx \mathbf{X} \mathbf{X}^\top$,

$$\mathbf{K} \approx \mathbf{X}(\mathbf{H}\mathbf{X})^\top \left[(\mathbf{H}\mathbf{X})(\mathbf{H}\mathbf{X})^\top + \mathbf{R} \right]^{-1},$$

which corresponds to the following covariance approximations:

$$\mathbf{P}^f \mathbf{H}^\top \approx \mathbf{X}(\mathbf{H}\mathbf{X})^\top = \frac{1}{N_e - 1} \sum_{i=1}^{N_e} (\mathbf{x}_i^f - \bar{\mathbf{x}}^f)(\mathbf{H}\mathbf{x}_i^f - \mathbf{H}\bar{\mathbf{x}}^f)^\top$$

$$\mathbf{H} \mathbf{P}^f \mathbf{H}^\top \approx \frac{1}{N_e - 1} \sum_{i=1}^{N_e} (\mathbf{H}\mathbf{x}_i^f - \mathbf{H}\bar{\mathbf{x}}^f)(\mathbf{H}\mathbf{x}_i^f - \mathbf{H}\bar{\mathbf{x}}^f)^\top.$$

Analysis covariance I

- We also **require** the **ensemble-based analysis covariance matrix** $\hat{\mathbf{P}}^a = \mathbf{X}^a \mathbf{X}^{a\top}$ **equal to** the one provided by Kalman filter. Hence,

$$\begin{aligned}\mathbf{X}^a \mathbf{X}^{a\top} &= (\mathbf{I} - \mathbf{K}\mathbf{H})\mathbf{P}^f = \mathbf{P}^f - \mathbf{P}^f \mathbf{H}^\top (\mathbf{H}\mathbf{P}^f \mathbf{H}^\top + \mathbf{R})^{-1} \mathbf{H}\mathbf{P}^f \\ &\approx \mathbf{X}\mathbf{X}^\top - \mathbf{X}\mathbf{X}^\top \mathbf{H}^\top (\mathbf{H}\mathbf{X}\mathbf{X}^\top \mathbf{H}^\top + \mathbf{R})^{-1} \mathbf{H}\mathbf{X}\mathbf{X}^\top \\ &= \mathbf{X} \left[\mathbf{I} - \mathbf{X}^\top \mathbf{H}^\top (\mathbf{H}\mathbf{X}\mathbf{X}^\top \mathbf{H}^\top + \mathbf{R})^{-1} \mathbf{H}\mathbf{X} \right] \mathbf{X}^\top.\end{aligned}$$

- This relation is satisfied if $\mathbf{X}^a = \mathbf{X}\mathbf{F}$ such that  *transformation matrix*

$$\mathbf{F}\mathbf{F}^\top = \mathbf{I} - \mathbf{X}^\top \mathbf{H}^\top (\mathbf{H}\mathbf{X}\mathbf{X}^\top \mathbf{H}^\top + \mathbf{R})^{-1} \mathbf{H}\mathbf{X}.$$

- Remark:** We say that $\mathbf{F}$ is the **square root** of the matrix on the right-hand side.

Analysis covariance II

- From the expression $\mathbf{X}^a = \mathbf{X}\mathbf{F}$, it is clear that the columns of $\mathbf{F}$ are **coefficients for linear combinations of the forecast ensemble perturbations** (see slide 30 for more details).
- Q

 What are the dimensions of the matrix $\mathbf{F}$? $\begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$
 $N_e \times N_e$
- The transformation matrix $\mathbf{F}$ is **not unique**: multiplying it by an orthogonal matrix $\mathbf{V}$ leaves $\mathbf{X}^a \mathbf{X}^{a\top}$ unchanged. $\mathbf{F}' = \mathbf{F}\mathbf{V}, \mathbf{F}'\mathbf{F}'^\top = \mathbf{F}\mathbf{V}(\mathbf{F}\mathbf{V})^\top = \mathbf{F}\mathbf{V}\mathbf{V}^\top\mathbf{F}^\top = \mathbf{F}\mathbf{F}^\top$
- E

 Try to convince yourself of this property.
- The fact that $\mathbf{F}$ is not unique implies there exist many different ways to enforce the consistency of the KF update.
 - Different choices lead to different algorithms (to be discussed in future lectures).

Computational considerations I

- From the expression

$$\mathbf{F}\mathbf{F}^\top = \mathbf{I} - \mathbf{X}^\top \mathbf{H}^\top (\mathbf{H}\mathbf{X}\mathbf{X}^\top \mathbf{H}^\top + \mathbf{R})^{-1} \mathbf{H}\mathbf{X},$$

it is clear that the **largest computational costs** will be associated with **inverting the matrix $\mathbf{H}\mathbf{X}\mathbf{X}^\top \mathbf{H}^\top + \mathbf{R}$** .

- Q What are the dimensions of this matrix?

$$N_y \times N_y$$

$$N_x \approx 10^9$$

$$N_y \approx 10^6 - 10^7$$

$$N_e \approx 10 - 100.$$

- E, WB Use the SMW identity

$$(\mathbf{A} + \mathbf{UCV})^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{U}(\mathbf{C}^{-1} + \mathbf{VA}^{-1}\mathbf{U})^{-1}\mathbf{VA}^{-1}.$$

to show that

$$\mathbf{I} - \mathbf{X}^\top \mathbf{H}^\top (\mathbf{H}\mathbf{X}\mathbf{X}^\top \mathbf{H}^\top + \mathbf{R})^{-1} \mathbf{H}\mathbf{X} = (\mathbf{I} + \mathbf{X}^\top \mathbf{H}^\top \mathbf{R}^{-1} \mathbf{H}\mathbf{X})^{-1}.$$

Computational considerations II

- Q *What is the practical significance of the equality*

$$\mathbf{I} - \mathbf{X}^\top \mathbf{H}^\top (\mathbf{H} \mathbf{X} \mathbf{X}^\top \mathbf{H}^\top + \mathbf{R})^{-1} \mathbf{H} \mathbf{X} = (\mathbf{I} + \mathbf{X}^\top \mathbf{H}^\top \mathbf{R}^{-1} \mathbf{H} \mathbf{X})^{-1} ? \quad N_e \times N_e$$

- This identity makes it clear that $\mathbf{F} \mathbf{F}^\top = (\mathbf{I} + \mathbf{X}^\top \mathbf{H}^\top \mathbf{R}^{-1} \mathbf{H} \mathbf{X})^{-1}$. But how can we **calculate** $\mathbf{F}$?
- A common strategy is to perform an **eigendecomposition** of $(\mathbf{I} + \mathbf{X}^\top \mathbf{H}^\top \mathbf{R}^{-1} \mathbf{H} \mathbf{X})^{-1}$.
 - ① $\mathbf{A} := \mathbf{I} + \mathbf{X}^\top \mathbf{H}^\top \mathbf{R}^{-1} \mathbf{H} \mathbf{X} = \mathbf{E} \mathbf{\Lambda} \mathbf{E}^\top$. $\mathbf{E}$ is an orthogonal matrix comprised of the eigenvectors of $\mathbf{A}$, while $\mathbf{\Lambda}$ is a diagonal matrix with the eigenvalues of $\mathbf{A}$.
 - ② $\mathbf{A}^{-1} = \mathbf{E} \mathbf{\Lambda}^{-1} \mathbf{E}^\top$. WB *Check this.*
 - ③ $\mathbf{F} = \mathbf{E} \mathbf{\Lambda}^{-1/2} \mathbf{E}^\top$. WB *Check this.*

Full analysis update

- The **final step** in deterministic EnKFs is to combine the analysis mean $\overline{\mathbf{X}}^a$ and perturbation matrix $\mathbf{X}^a$. $= \overline{\mathbf{X}}^a + \mathbf{F}$
- Similar to the stochastic EnKF, we write down an update equation for each forecast ensemble member as follows:

$$\mathbf{X}_i^a = \overline{\mathbf{X}}^a + \sqrt{N_e - 1} \mathbf{X}^a[:, i] \quad (4)$$

- Q Can you verify this result? Why did we multiply $\mathbf{X}^a[:, i]$ by $\sqrt{N_e - 1}$?
- Hence, the **general procedure** for all deterministic EnKFs is:
 - 1 Transform the forecast ensemble mean $\overline{\mathbf{X}}^f$ to $\overline{\mathbf{X}}^a$ using the KF's analysis mean equation.
 - 2 Transform the forecast ensemble perturbations $\mathbf{X}^f$ to $\mathbf{X}^a$ using the square-root matrix $\mathbf{F}$.
 - 3 Use (4) to recover the full analysis ensemble $\{\mathbf{X}_i^a\}_{i=1}^{N_e}$.

Nonlinear/non-Gaussian approximations

- Just like the stochastic EnKF, it is usually necessary to deal with **nonlinear/non-Gaussian situations** in reality.
- Within the deterministic EnKF framework, these are handled by **modifying** the **forecast mean update** as

$$\bar{\mathbf{X}}^a = \bar{\mathbf{X}}^f + \hat{\mathbf{K}} \left[\mathbf{y} - \mathbf{h}(\bar{\mathbf{X}}^f) \right] \rightarrow \text{mean eq. for the EKF}$$

and **further approximating** the **observation-space forecast perturbation matrix** $\mathbf{H}\mathbf{X}$ needed in $\hat{\mathbf{K}}$ as

$$\mathbf{H}\mathbf{X} = \text{matrix with columns } (N_e - 1)^{-1/2} \left[\mathbf{h}(\mathbf{X}_i^f) - \overline{\mathbf{h}(\mathbf{X}_i^f)} \right].$$

- While this strategy allows for practical implementations, it is **suboptimal**. Q Why?